

Organisation of the examination

All candidates must have access to the materials required for Question 1 throughout the whole period of the examination.

For Question 3, **each candidate must have sole access to a microscope for an uninterrupted period of 1 hour 15 minutes only**. Two candidates are not permitted to share the same microscope at the same time. Half of the candidates will have access to a microscope for a period of 1 hour 15 minutes from the start of the examination. These candidates should be advised to start with Question 3. Once candidates have finished working with the microscope they can move on to Question 1 or 2.

The other candidates should be advised to start with Question 1. After 1 hour 15 minutes, these candidates should be given access to the microscope and advised to move on to Question 3 as soon as possible.

Candidates should be advised that they can answer any part of the examination paper throughout the whole period of the examination. They will only have access to the microscope for Question 3 during the specified period of 1 hour 15 minutes.

Access arrangements to microscopes, including advice on which question to start with and timings, must be explained clearly to candidates before the start of the examination.

Question 1

Each candidate will require:

materials and apparatus for each candidate	quantity	✓
[MH][HH][C] 1.0% invertase solution in a small beaker, labelled E , provided at room temperature (see Preparation of materials)	at least 20 cm ³	
[MH][HH][C] 0.1% invertase solution in a small beaker, labelled H , provided at room temperature (see Preparation of materials)	at least 40 cm ³	
[MH][N] Benedict's solution (qualitative) in a beaker or container, labelled B , provided at room temperature	at least 50 cm ³	
5% sucrose solution in a small beaker, labelled S , provided at room temperature	at least 30 cm ³	
Distilled water, in a beaker or container, labelled W , provided at room temperature	at least 150 cm ³	
10 cm ³ syringe, with the means to wash it out	1	
5 cm ³ syringes, with the means to wash them out	3	
1 cm ³ syringe, with the means to wash it out	1	
Beakers, capacity 50–100 cm ³	5	
Bunsen burner, tripod, gauze, bench mat and lighter or matches	1	
Beaker, capacity approximately 400 cm ³ , with approximately 200 cm ³ of water at 30–40 °C, suitable for heating as a water-bath	1	
Test-tubes, small, capacity approximately 20–30 cm ³	6	
Test-tube holder	1	
Test-tube rack to hold 6 small test-tubes	1	
Glass rod	1	
Thermometer, –10 °C to +110 °C	1	
Container with approximately 200 cm ³ tap water, labelled For washing	1	
Container, capacity approximately 300 cm ³ , labelled For waste	1	
Paper towels	8	
Glass marker pen, permanent	1	
Stop-clock or timer showing seconds	1	
Suitable eye protection	1	

It is advisable to wear suitable eye protection when handling chemicals.

Gloves provided at teacher's bench

Preparation of materials

Sucrose solution, **S**, may be prepared the day before the exam.

The container should be covered and put in a refrigerator overnight but removed to reach room temperature before the start of the exam.

E, 1.0% invertase solution and **H**, 0.1% invertase solution

- prepared using invertase powder (Carolina)
- **E** and **H** must be prepared immediately before the exam and be at room temperature

S, 5% sucrose solution

This is prepared by putting 5 g of sucrose into 80 cm³ of warm distilled water and mixing well. Make up to 100 cm³ with distilled water.

Official (Closed)/ Sensitive (High)

Question 3

Each candidate must have sole, uninterrupted use of a microscope for 1 hour 15 minutes only.

For each candidate:

- the microscope must be set up to the low-power objective lens
- the slide must **not** be left on the stage of the microscope.

Preparation List

materials and apparatus for each candidate	quantity	✓
Microscope with an eyepiece graticule	1 between 2	
<i>Zea mays</i> root (ts) slide labelled ' J1 '	1 between 2	
Sweet potato cylinder in a damp paper towel, in a Petri dish base (J2)	1	
Iodine (in potassium iodide) solution, in a suitable container labelled iodine and provided with a dropper	at least 10 cm ³	
Distilled water, in a capped specimen tube labelled (W)	At least 10 cm ³	
Microscope slide and coverslip	1	
Mounted needle	1	
Stage micrometer	1	
Scalpel, sharp	1	
Ruler, clear plastic	1	
Pasteur pipette	1	
Blunt forceps	1	
White tile	1	
Base of petri dish	1	
Marker	1	
Safety goggles	1 pair	
Paper towel	8	

Cylinder of tissue from sweet potato tuber, in a Petri dish labelled **J2**

Use a cork borer with a diameter of about 13 mm (e.g. a number 7 cork borer) to obtain cylinders of tissue from sweet potato tubers.

Cylinders should be trimmed on both sides (o remove skin) to a length of at least 2 cm and wrapped in a damp paper towel.

Iodine (in potassium iodide) solution, in a suitable container labelled **iodine**

Standard iodine solution suitable for testing food for starch can be used, e.g. commercially prepared 0.1N/0.05 moldm⁻³ iodine solution